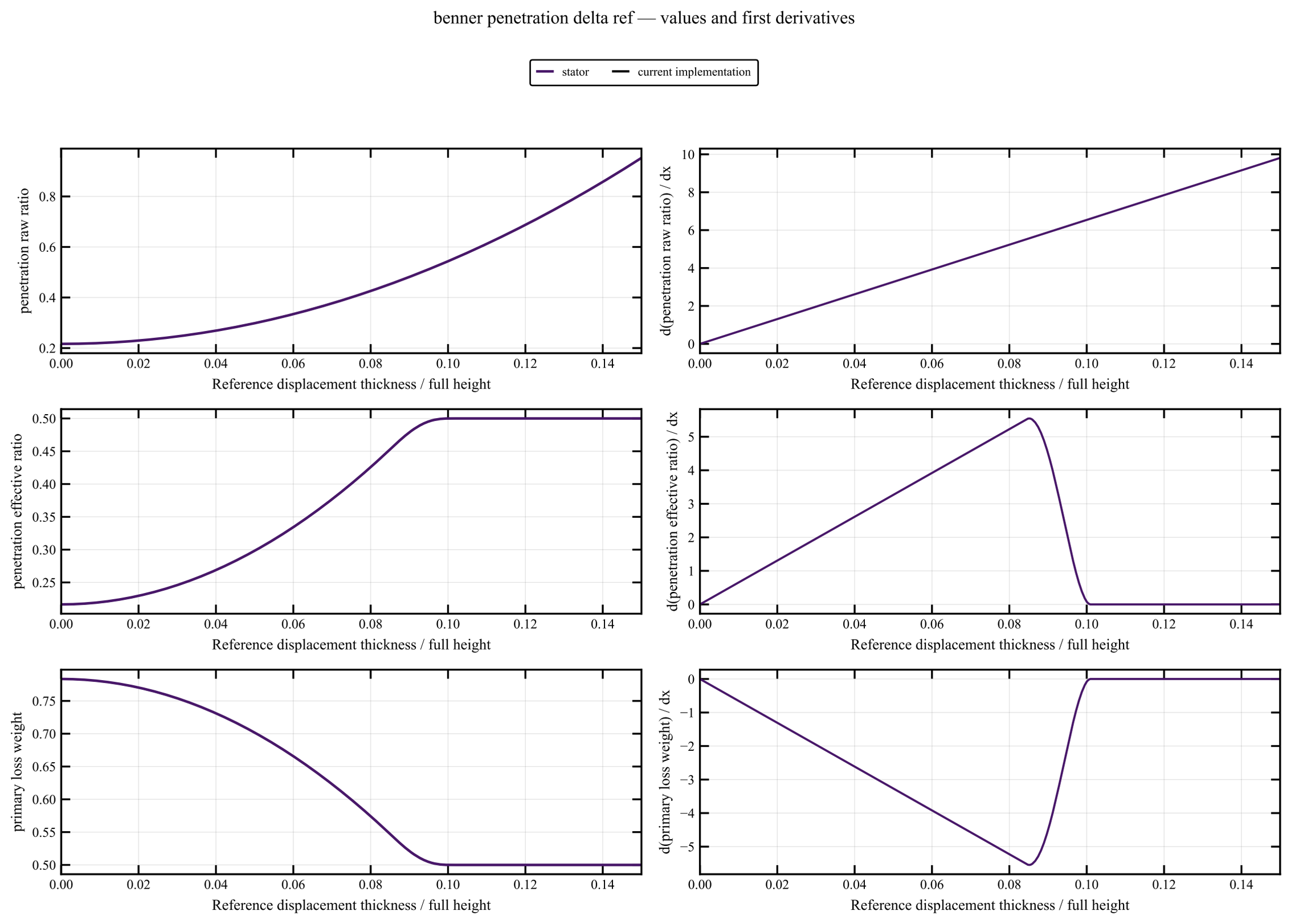benner penetration delta ref — values and first derivatives